

3rd

```
medikondarakesh@Medikondas-MacBook-Air ~ % nano fork_demo.c
medikondarakesh@Medikondas-MacBook-Air ~ % gcc fork_demo.c -o fork_demo
./fork_demo
Parent Process
PID : 5158
Child Process
PID : 5159
PPID: 5158
medikondarakesh@Medikondas-MacBook-Air ~ %
```

UW PICO 5.09

File: fork_demo.c

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid = fork();

    if (pid == 0) {
        printf("Child Process\n");
        printf("PID : %d\n", getpid());
        printf("PPID: %d\n", getppid());
        sleep(10);
    } else {
        printf("Parent Process\n");
        printf("PID : %d\n", getpid());
        wait(NULL);
    }

    return 0;
}
```

4th

```
UW PICO 5.09 File: multi_fork.c Modif

#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int i;

    // Loop to create 3 child processes sequentially
    for (i = 0; i < 3; i++)
    {
        if (fork() == 0)
        {
            // Executed by child process
            printf("Child %d PID=%d\n", i + 1, getpid());
            return 0; // Terminate child immediately so it does not continue the loop
        }
    }

    // Parent waits for all 3 children to finish execution
    for (i = 0; i < 3; i++)
    {
        wait(NULL);
    }

    printf("All children completed.\n");
    return 0;
}

G Get Help      O WriteOut      R Read File      Y Prev Pg      K Cut Text      C Cur Pos
X Exit          J Justify        W Where is      V Next Pg      U UnCut Text    T To Spell
```

```
medikondarakesh@Medikondas-MacBook-Air ~ % nano multi_fork.c
medikondarakesh@Medikondas-MacBook-Air ~ % gcc multi_fork.c -o multi_fork
[./multi_fork
Child 1 PID=5177
Child 2 PID=5178
Child 3 PID=5179
All children completed.
medikondarakesh@Medikondas-MacBook-Air ~ %
```